FLORIA TSUI

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EDUCATION

BACHELOR of SCIENCE: Computer Science & Political Science

Brown University | 2019-2023 | GPA: 3.88

Past courses: Fundamentals of Computer Systems, Object Oriented Programming, Functional Programming, Algorithms and Data Structures, Deep Learning, Software Security and Exploitation

Current courses: Operating Systems, Computer Systems Security, Discrete Structures and Probability

EXPERIENCE

TEACHING ASSISTANT

Introduction to Computer Systems | Fall 2020
Computing Foundations: Data | Spring 2020

- 1 of 14 TAs hired as a freshman out of 100+ applicants
- Graded homework assignments for 150+ students and held office hours, helping students debug code in Pyret and Python (4 hours / week)
- Revamped and implemented 3 course projects: analyzing large datasets, building a maze game using lists in Pyret, and generating a book recommendation site using hashtables in Python

RESEARCH ASSISTANT

Department of Sociology | Summer 2020

- Created a dictionary of technocratic terms (100+) based on news articles and journals
- Developed Python code that would use the developed dictionary to analyze the frequency and use of technocratic and expertise based language in discussions about climate change

FINANCE FELLOW

Gretchen Driskell for Congress | Summer 2020

- Directly assisted Finance Director with fundraising events by finding and analyzing prospective attendees, designing invitations, and aiding in recruitment
- Helped find prospective donors for interns to research and analyzed other candidates' fundraising patterns
- Raised $200+ through my own fundraising efforts

SUMMER FIELD INTERN

Generation Rising and Everyday People PAC | Summer 2020

- Phone banked to Florida and Arizona, making over 300 calls in 2 weeks (9 hours / week)
- Textbanked to Pennsylvania, texting 3000+ people to register people to vote

PROJECTS

SHARDING

C++ | Spring 2020

- Created a sharded, distributed key-value storage system using gRPCs and protobufs
- Implemented a "shardmaster" that assigns shards to key-value servers and updates the shard distributions when servers join or leave
- Constructed a key-value server, where users can get, put, and append and transfer key-value pairs depending on the shard configuration

SEARCH

Scala | Spring 2020

- Wrote a search engine for users to query a corpus of Wikipedia pages and returns the most relevant pages
- Developed an Indexer that processes the text of each page that stores each word and its frequency into hashtables, using Google's page rank algorithm (< 1 min)
- Developed a Querier that processes a free text query from the user, calculates the relevance score, and returns the pages associated with the query (< 1 sec)

WEENSYOS

C++ | Spring 2020

- Separated the address space of the operating system for kernel and process isolation
- Implemented the basics of virtual memory architecture using pagetables
- Wrote the system calls fork and exit

VUNMO

C++ | Spring 2020

- Implemented a concurrent banking service, similar to the application Venmo
- Created a synchronized queue using condition variables for a pool of "worker" threads that would handle user requests
- Developed a multi-threaded server that accepts clients, recieves and reads client requests, and procesess each request

SKILLS & INTERESTS

PROGRAMMING LANGUAGES- Python, C, Java, Scala, C++

DESIGN- Adobe Photoshop & Illustrator

- Staff Illustrator for the College Hill Independent, contributing 4 illustrations (5 hours / week)

PIANO

- Played for 10+ years & performed at Carnegie Hall